PROJECT APPROACH

Title: AI Assisted Data Visualization for Business Decision Support Systems

Domain: Business analytics and decision support systems.

Datasets:

1. Superstore Sales Dataset:

The **Superstore Sales Dataset** is a retail business dataset that contains transactional information such as customer details, product categories, sales, profit, discounts, quantity, shipping mode, region, state, and order details. It is used to analyze business performance, identify sales trends, evaluate profitability, perform customer segmentation, and support AI-assisted business decision-making through interactive data visualization.

Dataset URL

2. Global Superstore Dataset:

The **Global Superstore Dataset** is a multinational retail business dataset that contains transactional information from multiple countries, including customer details, product categories, sales, profit, shipping, market, and regional data. It is used to analyze global business performance, compare regional markets, identify sales trends, evaluate profitability, and support AI-driven business decision-making through interactive data visualization.

Dataset URL

3. Brazilian E-commerce Public Dataset by Olist:

The **Brazilian E-Commerce Public Dataset (Olist)** is a real-world e-commerce dataset that contains information about online orders, customers, products, sellers, payments, deliveries, and customer reviews. It is used to analyze customer purchasing behavior, delivery performance, sales trends, payment patterns, and customer satisfaction, enabling AI-assisted business decision-making through interactive data visualization.

Dataset URL

Abstract:

The rapid growth of business data has created a need for intelligent systems capable of transforming raw information into meaningful insights for effective decision-making. This project, "AI-Assisted Data Visualization for Business Decision Support Systems," aims to develop an intelligent analytics platform that combines data science, machine learning, and interactive visualization techniques to support business decision-making processes. The project utilizes benchmark business datasets, including the Superstore Sales Dataset, Global Superstore Dataset, and Brazilian E-Commerce Public Dataset (Olist), to analyze sales performance, customer behavior, profitability, and regional business trends.

The proposed system follows a systematic data science pipeline consisting of data collection, preprocessing, exploratory data analysis (EDA), feature engineering, feature selection, machine learning model development, explainable artificial intelligence (XAI), and interactive dashboard visualization. Advanced machine learning algorithms such as Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost are employed to generate predictive business insights, while SHAP and LIME techniques provide transparent explanations for model predictions. An interactive dashboard developed using Streamlit enables users to visualize key performance indicators (KPIs), monitor business trends, and explore analytical results in real time.

The primary objective of this project is to bridge the gap between complex data analytics and business decision-making by providing an AI-assisted visualization platform that delivers accurate predictions, meaningful insights, and actionable recommendations. The proposed system is expected to enhance organizational decision-making, improve business performance analysis, and demonstrate the practical application of artificial intelligence and data visualization techniques in modern business intelligence systems.

Data Analytics:

Exploratory Data Analysis (EDA Workflow Diagram):

Raw Business Datasets

|

▼

Dataset Overview

|

▼

Business Understanding

|

▼

Data Quality Assessment

|

▼

Descriptive Statistics

|

▼

Missing Values Analysis

|

▼

Duplicate Records

|

▼

Outlier Detection

|

▼

Univariate Analysis



Preprocessing:

1. Data Cleaning

- Missing Value Handling
- Duplicate Removal
- Outlier Detection & Treatment

2. Data Encoding

- Label Encoding
- One-Hot Encoding

3. Data Scaling & Transformation

- Feature Scaling
- Data Normalization
- Data Transformation

4. Feature Preparation

- Feature Engineering

Machine Learning Models:

Tree-Based Models:

- Decision Tree

Ensemble Learning Models:

- Random Forest
- XGBoost
- LightGBM
- CatBoost

Explainable AI:

Explainable AI Techniques:

- SHAP (Shapley Additive exPlantations)
- LIME (Local Interpretable Model-Agnostic Explanations)

Visualization:

Statistical Visualizations:

- Heatmap
- Boxplot
- Scatter Plot
- Correlation Matrix

Business Visualization:

- Sales and Trend Analysis
- Profit Analysis
- Customer Segmentation Charts
- Regional Performance Analysis
- Product Category Analysis

Interactive Dashboard:

- KPI Cards
- Interactive Dashboard (Streamlit)
- Feature Importance

Comparative Analysis:

Comparing:

- Decision Tree
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1 – Score
- ROC – AUC
- MAE (Mean Absolute Error)
- RMSE (Root Mean Square Error)

Research Pipeline:

Business Datasets

|

▼

Data Collection

|

▼

Data Preprocessing

|

▼

Exploratory Data Analysis (EDA)

|

▼

Feature Engineering

|

▼

Feature Selection

|

▼

Machine Learning Models

|

▼

Explainable AI (SHAP & LIME)

|

▼

Model Evaluation

|

▼

Comparative Analysis

|

▼

Interactive Business Dashboard

|

▼

Business Insights & Decision Support

System Architecture:

Frontend:

- Streamlit

Backend:

- Python

Machine Learning Engine:

- Scikit – Learn
- XGBoost
- LightGBM
- CatBoost

Explainability Module:

- SHAP
- LIME

Visualization Engine:

- Plotly
- Matlplotlib

Deployment Architecture Diagram:

